POF-Based Curvature Closed-Loop PI Control for a Pneumatic Soft Hand Exoskeleton

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Abstract—**Accurate and stable motion control in pneumatically actuated soft hand exoskeletons is limited by nonlinear fluid–structure interactions and the difficulty of obtaining reliable feedback without compromising compliance. This work presents the integration of a polymer optical fiber (POF) curvature sensor into a closed-loop control architecture for a soft hand exoskeleton, enabling direct curvature regulation through a low computational cost PI-based controller. By using curvature as the controlled variable, the proposed framework avoids complex pressure modeling while maintaining the intrinsic compliance of the system. Experimental validation is conducted using step-based curvature references representative of rehabilitation tasks, and performance is assessed through standard tracking metrics. The results show consistent tracking behavior with stable and well-damped responses, highlighting the effectiveness of curvature-based feedback for practical soft hand exoskeleton control.**

Index Terms—**Polymer optical fiber sensors, hand exoskeleton, pneumatic actuators, closed-loop PI control.**

I. INTRODUCTION

Stroke is a major cause of long-term disability and often leads to motor impairments in the upper extremities, particularly affecting hand function [1]. The human hand is essential for interacting with the environment, in particular for performing activities of daily living (ADLs) [2]. The impairment of hand function highlights the importance of using rehabilitation strategies to restore hand motor function, with the goal of improving the affected person's quality of life. One of the most widely used therapeutic strategies to support motor recovery is repetitive task practice, where intensive, task-specific training is used to drive neuroplastic changes and restore mobility [3]. In fact, these clinical needs have led to increasing interest in robotic devices to support users during hand rehabilitation and everyday tasks.

In recent years, upper limb exoskeletons have emerged as promising assistive and rehabilitation tools to deliver consistent training intensity, increase patient engagement, and support rehabilitation tasks [4]. Most devices fall into two main categories: rigid and soft robotic exoskeletons. Rigid exoskeletons provide higher force transmission, improved grip stability, and greater control over individual degrees of freedom (DoFs) [5]. However, these devices are often heavy and bulky, restricting their applicability in everyday environments [2], [3], [5].

In contrast, soft robotic exoskeletons use highly deformable and compliant materials to preserve the user's safety, dexterity, and comfort [3], [5]. These characteristics make soft robotic exoskeletons suitable as wearable assistive systems to assist in the performance of ADLs [6], [7].

Within soft robotics, pneumatic actuation is a widely used approach due to its high compliance for rehabilitation applications [6]. This actuation is a commonly used technology for its lightweight components and straightforward implementation. Soft Pneumatic Actuators (SPAs) are flexible structures whose material composition and geometry vary between different designs, but all depend on internal pressure to generate deformation [8]. In SPAs, applying pressure or vacuum to the internal chamber produces bending or extension movements, establishing a relationship between pressure and deformation [3]. This pressure–curvature coupling has motivated the development of closed-loop control strategies that aim not only to regulate pressure but also to achieve accurate positional behavior through model-based or empirical mappings [9].

In parallel to pneumatic actuation, the other major branch of soft robotics for hand assistance is actuation by using tendon-driven mechanisms, where flexible cables or tendons transmit force to the fingers and allow the user to achieve different grips. These gloves provide high compliance and user comfort while preserving realistic finger trajectories, making them useful for assistive applications and rehabilitation [4]. Nevertheless, pneumatic actuation provides unmatched intrinsic safety, as the deformation of the actuator directly results from the pressurization of soft material [3], [5]. To effectively use pneumatically actuated soft hand exoskeletons in functional tasks, their deformation must be monitored with sensing technologies that preserve compliance. Accordingly, reliable curvature-related feedback is essential for closed-loop control of SPAs [8].

Considering the flexible nature of soft actuators, these sensors must be flexible to avoid restricting the deformation of the soft structure. Across recent developments in pneumatic hand exoskeletons, several sensing modalities have been integrated directly into SPAs, including resistive and piezoresistive strain sensors, capacitive stretch sensors, magnetic deformation sensors, acoustic waveguides, and soft pneumatic chambers that function as intrinsic proprioceptive sensors [8].

Polymer Optical Fiber (POF) sensors are highly suitable for this purpose, as they combine low stiffness, weight, impact resistance, and low manufacturing cost [10]. In addition, POF-based sensors exhibit immunity to electromagnetic interference (EMI), a major advantage in pneumatic exoskeletons, where solenoid valves, motors, and switching electronics introduce significant noise that affects electronic sensors [10], [11]. The principle of POF sensors relies on variations in transmitted light intensity caused by the bending of the fiber. This facilitates direct measurement of curvature and joint trajectories, making POF sensors effective for tracking finger motion [12]. Finally, once reliable curvature measurements are available, they can be directly integrated into closed-loop control strategies to drive actuator behavior, where POF-based sensing provides accurate, deformation-compatible curvature feedback while fully complying with soft pneumatic structures [10], [11].

A review of the control strategies reported in the literature reveals a consistent progression toward achieving accurate and robust control of SPAs. The survey explores a wide range of techniques, including nonlinear feedback control [9], dual-valve and multivalve configurations for improved pressure regulation [13], model-based controllers for pressure tracking [14], and hybrid feedforward–feedback architectures designed to compensate for valve dynamics and system nonlinearities [8]. Although these approaches demonstrate significant advances in pressure control and dynamic response, most existing studies primarily regulate internal pressure rather than directly controlling the curvature or functional motion of the actuator and also tend to rely on proportional–integral–derivative (PID) controllers implemented using on–off valve actuation.

This paper presents the integration of a POF curvature sensors into a closed-loop control architecture for a pneumatic soft hand exoskeleton. The proposed system combines compliant curvature sensing with a low computational cost PI-based control strategy, enabling stable, repeatable, and well-damped motion feasible for assistive and rehabilitation-oriented tasks. By directly regulating finger curvature, the control framework avoids complex pressure or force modeling while preserving the inherent compliance of the soft structure.

The remainder of this paper is organized as follows: Section II describes the overall system architecture, including the instrumentation pipeline, sensor integration, the experimental setup and performance metrics. Section III presents the system identification results and the formulation of the proposed curvature-based control framework, followed by experimental validation under representative step-based reference trajectories and the corresponding closed-loop performance metrics. Section IV discusses the observed control behavior and its practical implications for soft hand exoskeleton operation. Finally, Section V summarizes the main conclusions and outlines directions for future work.

II. METHODOLOGY

A. System Description

1) Hand Exoskeleton: The soft hand exoskeleton used in this study is a commercial device for stroke rehabilitation (Syrebo, China). The original system includes two air pumps and a solenoid valve for airflow regulation. The system was modified by using two diaphragm pumps of the D2028 series (Forever/Anporn, China), and array of six three-way, two-position (3/2) solenoid valves, each with one inlet, one outlet, and one exhaust port (Adafruit Industries LLC, USA) to control the air flow to each finger. As for the electronics, MOSFET transistors (SI9956DY, Vishay Siliconix, USA) and optocouplers (MCT62S, Onsemi, USA) were implemented to control the pneumatic system. The processing and control of the device is performed by a single-board computer (Raspberry Pi 4 Model B) running Ubuntu Server with the Robot Operating System 2 (ROS 2). All mechanical and electronic components were housed in a custom enclosure control box.

2) POF Sensors: POF sensors have been manufactured as shown in [12]. In addition, Thermoplastic Polyurethane (TPU) 3D-printed pieces are used to support and guide the fiber around the fingers. The optical fiber used to fabricate the POF sensor is the SH4001 (Mitsubishi Chemical Corporation, USA), a Super Eska Simplex Optical Fiber Cable with a 2.2 mm OD Polyethylene Jacket. The fiber is made of a polymethyl-methacrylate resin core with a fluorinated polymer cladding and a core diameter ranging from 920 to 1040 µm. To improve the sensitivity of the POF-based sensor, a sensitive area was fabricated by polishing a part of the fiber core and cladding, also it was applied a thermal treatment as is described in [10], [11]. Finally, the setup requires light-emitting diodes (IF-E97, Industrial Fiber Optics, USA) and phototransistors (IF-D93, Industrial Fiber Optics, USA). Then, the signal is read by a 32-bit resolution analog-to-digital converter (ADC) board (ADS1263, Waveshare, China), and the resulting signals were used as feedback for identification and control.

B. Identification and Control

Given the nonlinear and pressure–dependent mechanical behavior of SPAs, obtaining a reliable control model requires combining experimental characterization with simplified dynamic representations that remain feasible for real-time implementation. In this work, the actuator–sensor pair (SPA + POF) was modeled through a data-driven identification process, which has also been successfully applied to SPAs in previous studies [15], [14].

1) Experimental Setup and System Identification Protocol: Figure 1 shows the complete experimental setup, including the POF sensor placement and the pneumatic actuation hardware. The hand exoskeleton was mounted on a 3D-printed support to provide a stable base to maintain the full range of fingers motion. For system identification, the highlighted selector valve (Fig. 1) was driven by a PWM signal to alternate between positive and negative pressure inputs. A complete sweep of the duty-cycle (0–100% and 100%-0%) was applied to adequately excite the system moving, and the resulting response to curvature was measured through POF sensors. To ensure that each operating point was properly excited, the PWM cycle was incremented in steps of 20%, and each step was held for 60 seconds. This procedure captured the pressure-curvature dynamics of the actuator under both inflation and deflation conditions.

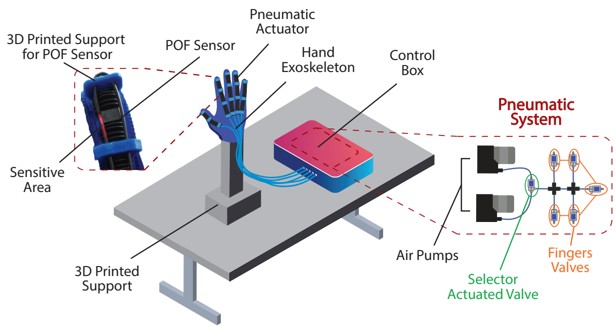

Fig. 1. Experimental setup for system identification.

2) Control Strategy: The gains were calculated using a pole placement method by assigning the closed-loop dynamics to a double pole located at $-\alpha$. This yields the analytical expressions $K_c = (2\alpha\tau - 1)/K$ and $T_i = (2\alpha\tau - 1)/(\tau\alpha^2)$.

Where α defines the desired closed-loop pole location, τ and K correspond to the time constant and static gain of the actuator model, respectively, K_c is the proportional gain, and T_i is the integral time constant of the PI controller. The resulting controller equation is then given by:

$$C(s) = K_c + \frac{K_c}{sT_i}, \tag{1}$$

3) Implementation: The continuous-time PI controller was implemented in ROS 2 as a discrete incremental algorithm executed at a sampling period T_s. Following the digital implementation of PID controllers based on backward-rectangular integration [16], the PI law was written as

$$u[k] = u[k-1] + q_0\, e[k] + q_1\, e[k-1], \tag{2}$$

where $e[k]$ is the curvature tracking error and $u[k]$ is the internal control action. The incremental gains are $q_0 = K_c$ and $q_1 = \frac{K_c}{T_i}T_s - K_c$. This expression is implemented in the control node, the previous control action $u[k-1]$ is stored, and the error vector $\begin{bmatrix} e[k] & e[k-1] \end{bmatrix}$ is updated at each loop iteration.

The internal control signal output $u[k]$ is signed, so it is mapped to a non-signed PWM duty cycle used to command the selector valve. The neutral valve condition is centered at 50%, which corresponds to the natural resting curvature of the soft hand exoskeleton when unpowered. Accordingly, the mapping is defined as $\mathrm{w}[k] = 50 + k_u u[k]$, where k_u is a scaling factor, and the resulting duty cycle is saturated within the interval $[0, 100]\%$. In this configuration, positive values of $u[k]$ increase the duty cycle above 50%, driving the actuator towards positive pressure (inflation), while negative values of $u[k]$ decrease it below 50%, driving the system towards vacuum (deflation). This symmetric mapping around 50% simplifies the controller design and allows the PI action to modulate both inflation and deflation using a single PWM output.

For each step segment, the closed-loop performance was quantified using the integral of absolute error (IAE), the rise time (T_r), and the total variation of the control input, TV. The IAE captures the accumulated tracking error over the duration of each step, providing a robust measure of overall regulation accuracy that is less sensitive to brief peaks than pointwise error metrics. The T_r was computed as the time required for the measured curvature to reach 90% of the commanded transition, thus characterizing the transient responsiveness of the pneumatic actuation. To assess the control activity, it TV was computed as the sum of the absolute sample-to-sample changes in the PWM duty cycle. The TV metric is commonly used to measure performance related to control effort. More aggressive control signals produce higher TV values, requiring more energy and potentially accelerating actuator wear [17]. Then, the metric is calculated as

$$TV = \sum_{i=1}^{N} |u(i+1) - u(i)| \tag{3}$$

the metrics reported below correspond to the mean and standard deviation obtained across all step segments and experiments.

III. RESULTS

In this work, the middle finger was used as the reference for feedback control, while the same control signal was applied to all fingers due to the shared pneumatic actuation. It was selected because of its central position. As a result, only the middle finger closes the feedback loop, whereas the remaining fingers operate in an open loop but follow the same command, producing coordinated hand motion with reduced sensing and control complexity.

A. System Identification

The data obtained by the POF curvature sensor were first filtered using an exponential weighted moving average (EWMA) filter with a λ of 0.2, which attenuates high-frequency noise while preserving the underlying dynamic behavior of the actuator. Figure 2 shows the filtered and normalized data obtained by the POF curvature sensor, as well as the step changes in the PWM. From this dataset, each PWM step was segmented and treated as an individual input–output pair, with the PWM command as the excitation signal and the middle-finger POF-based curvature as the measured output.

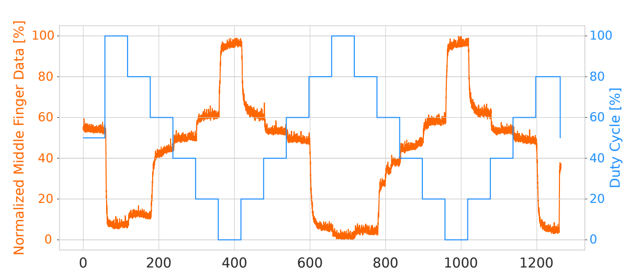

Fig. 2. Data from middle finger: PWM duty-cycle vs POF-based curvature.

To obtain models suitable for control design, it is important to note that PWM-driven 3/2 solenoid valves have nonlinearities such as dead zones, saturation near extreme duty cycles, and valve opening delays [18]. Than the analysis was restricted to the effective operating range between 20% and 80% duty cycle. So, step responses of 20–40%, 40–60%, and 60–80% bands were selected for both inflation and deflation, and each band was identified separately. Rather than aiming at a detailed nonlinear description of the system, the objective was to derive a low-order approximation that captures the dominant curvature dynamics. Based on the filtered input–output data, the actuator behavior in each band was approximated by a first-order model of the form:

$$G(s) = \frac{K}{\tau s + 1}, \tag{4}$$

where K and τ denote steady-state gain and time constant, respectively. The identified values of these parameters for each excitation band, as well as the overall mean model used for controller design, are summarized in Table I.

TABLE I
IDENTIFIED FIRST-ORDER AVERAGED MODELS FOR OPENING AND CLOSING BANDS AND THE OVERALL MEAN.

Direction	Duty band [%]	K	τ [s]
Opening	20–40	-1.69	6.81
Opening	40–60	-0.39	5.09
Opening	60–80	-0.24	4.02
Closing	80–60	-0.37	3.05
Closing	60–40	-0.57	5.37
Closing	40–20	-1.85	5.51

Figure 3 illustrates the identified dynamics for both closing and opening transitions. In the closing step Fig. 3a, the model captures the dominant transient and settling trend with a fit of 86.9%, whereas the opening step (Fig. 3b) has a slightly lower fit of 83.8%, reflecting the expected asymmetry between pressurization and depressurization in fluidic soft actuation. Overall, the obtained fits indicate that the identified model reproduces the dominant dynamics with adequate accuracy to support subsequent controller design and closed-loop analysis.

After first-order models were obtained for all excitation bands, the identified parameters were averaged within each band to obtain representative plants. These averaged models capture the main dynamic behavior relevant for curvature control and were used as the basis for the subsequent PI controller design.

B. Closed-Loop Control Performance

The PI controller, with gains $K_p = 15.58$ and $K_i = 6.34$, was tuned using an averaged model and implemented on the Raspberry Pi within the ROS 2 control pipeline. Figure 4 shows a representative closed-loop experiment, where the reference curvature profile is tracked by the measured POF-based curvature sensor.

To evaluate the closed-loop performance, three experiments were conducted in which the reference curvature was varied between 20% and 80%. The exoskeleton was driven through

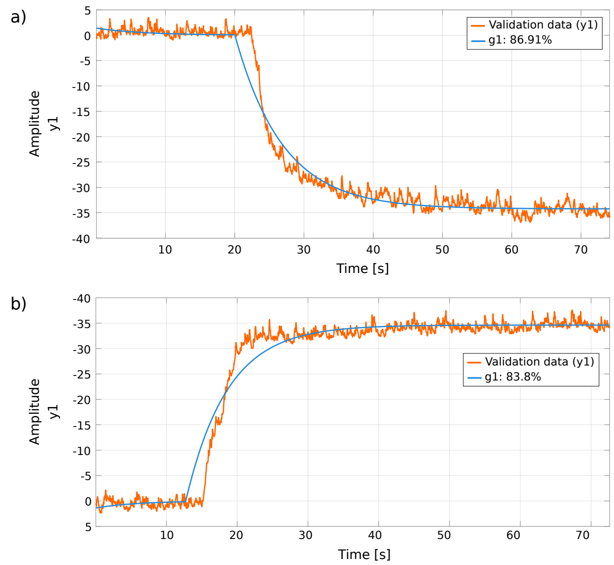

Fig. 3. a) Example of soft exoskeleton closing model and data comparative; b) Example of soft exoskeleton opening model and data comparative.

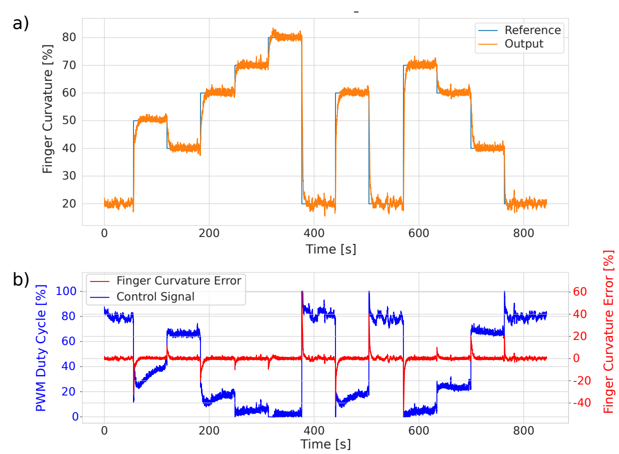

Fig. 4. a) Output vs Reference for different step values; b) PWM Control Signal and Error.

step commands of 20%, 40%, 50%, 60%, 70%, or 80% during both opening and closing motions, as shown in Figure 4a. Likewise, Figure 4b presents the control signal in PWM, aligned with the error value from the closed loop. Step references were adopted because they are commonly used for closed-loop performance assessment, allowing a direct evaluation of tracking behavior, transient response, and control effort.

TABLE II
CLOSED-LOOP PERFORMANCE METRICS.

Metric	Value
IAE	2.13 $\pm$0.58
T_r [s]	12.44 $\pm$3.42
TV	557.63 $\pm$135.00

As depicted in Table II, the low dispersion of the IAE indicates consistent tracking across repetitions, with most of the error concentrated around step transitions rather than during the steady-state plateaus. The calculation of the indexes Tr and TV indexes is more sensitive to noise in the system, this explain their greater standard deviations. This interpretation is consistent with standard control-performance practice, as is shown by Chen *et al.* [17], where IAE is used to quantify tracking accuracy, while the total variation TV of the control input reflects signal smoothness (control activity) and is expected to be more sensitive to noise and switching effects.

IV. DISCUSSION

The use of a single 3/2 valve imposes a trade-off between flow resolution and response speed inherent to PWM-modulated pneumatic supplies. In this experiment, this appears as a ripple in the response to curvature, which is consistent with the pressure ripple observed when regulating the pressure of the chamber using a single 3/2 on–off valve on the PneuSoRD platform [19]. Although PneuSoRD focuses on pressure control rather than curvature control, the same PWM-driven on–off actuation scheme underlies the curvature ripple observed in the experiments. While this ripple introduces small oscillations in curvature, it does not compromise closed-loop stability or steady-state tracking within the operating range considered.

In addition, Figure 4 illustrates a representative closed-loop trial from which the performance metrics in Table II were computed. In Fig. 4a, the output curvature follows the six reference levels with negligible overshoot and small steady-state offsets, while most of the tracking error is concentrated around the rising and falling edges after each step. This behavior is captured by the mean IAE of 2.13 ± 0.58, indicating that the accumulated error over each step remains modest once the new curvature plateau is reached.

According to the Fig. 4b, the corresponding PWM duty cycle exhibits saturation near the 0–100% limits as the controller compensates for valve nonlinearities and the soft-actuator dynamics. The T_r of 12.44 ± 3.42 s therefore reflects a practical compromise between fast transitions and smooth, well-damped motion, given the constraints of a single on–off valve and EWMA-filtered curvature feedback.

Furthermore, the variability in TV is expected since the total variation measures the control effort, which varies due to changes in the process gain for different regions (see Table I). Than, the lower the gain, the stronger the control effort. Accordingly, TV captures the overall control activity required to achieve robust curvature tracking, whereas the low IAE confirms accurate regulation once each plateau is reached.

Overall, these metrics confirm that the proposed controller is adequate for low-frequency, rehabilitation-oriented tasks, which is consistent with the design philosophy of soft hand exoskeletons, where movements are intentionally slow, safe, and repetitive rather than fast or highly precise. Such behavior is consistent with prior work showing that soft pneumatic gloves for ADL assistance prioritize comfort and controlled motion during grasping [6]. Moreover, SPAs inherently exhibit slow dynamic responses due to flow restrictions, viscoelastic material behavior [8], [18], which makes low-frequency curvature control not only acceptable but expected in rehabilitation scenarios. In this context, the controller's T_r and tracking accuracy fall within the typical operating regime of assistive soft robotic gloves, whereas achieving faster and higher-precision curvature variations required for hand gestures such as pinch or fine manipulation would need different actuation hardware or advanced control strategies.

V. CONCLUSIONS AND FUTURE WORK

The closed-loop experiments indicate that curvature regulation provides a reliable and repeatable way to command pneumatically actuated soft fingers for rehabilitation-oriented motions. Across step references between 20% and 80%, the controller achieved consistent transient behavior, with IAE and T_r aligned with the inherent bandwidth of pneumatic actuation. The TV measurement is consistent with the asymmetric behavior of inflation–deflation response and valve saturation rather than persistent steady-state inaccuracies. Overall, these results support curvature feedback as a meaningful control variable for soft hand assistance when smooth, low-frequency motions are prioritized.

Future work will focus on implementing alternative sensing modalities for curvature feedback, such as textile strain sensors, resistive flex sensors, and lightweight inertial units. This will allow benchmarking of sensing technologies to determine the most effective approach for real-time curvature estimation. The proposed controller will then be extended from whole-hand operation to independent control of each finger and to the regulation of intermediate hand-opening positions, enabling more flexible hand configurations, more accurate execution of grasp patterns, and more efficient hand gesture performance. Finally, preliminary validation of the device is expected to be carried out, first in healthy users and then in post-stroke patients, to determine whether the device contributes to improving patient performance.

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